

ENVIRONMENT AND ITS SEGMENTS

- Word “**Environment**” derived from French word “**Environner**” meaning **“To encircle or Surround”**.
- Simply be defined as the **surrounding of an organism** in which the **organism lives**.
- **Surrounding** includes things-
 - ✓ **Non-living (Abiotic)- Air, Water, Soil etc.**
 - ✓ **Living (Biotic)- all other living beings which organism comes into regular contact in its environment.**
- Absence of these organism meaningless to speak of **environment and vice versa**.
- In other words exists a **mutual interaction** between **every organism and its environment**.

Global Environment

Consist four segments-

1. Atmosphere
2. Hydrosphere
3. Lithosphere
4. Biosphere

1. ATMOSPHERE

- **Definitions:** “*The thin envelope of gases surrounding the earth* **Highly compressible**”
- **Absorb** most cosmic rays from outer space and a major **portion of electromagnetic radiation (EMR)** from Sun and transmits only **near UV, visible & IR radiation (300-2500nm)** and **radio wave** while filtering out **harmful UV radiation below 300nm**.
- **Atmosphere is bound to earth by gravity.**
- **Total mass of Atmosphere:** 5×10^5 **tones** which is **1 millionth** of Earth's total mass.
- **Density decreases rapidly with height**

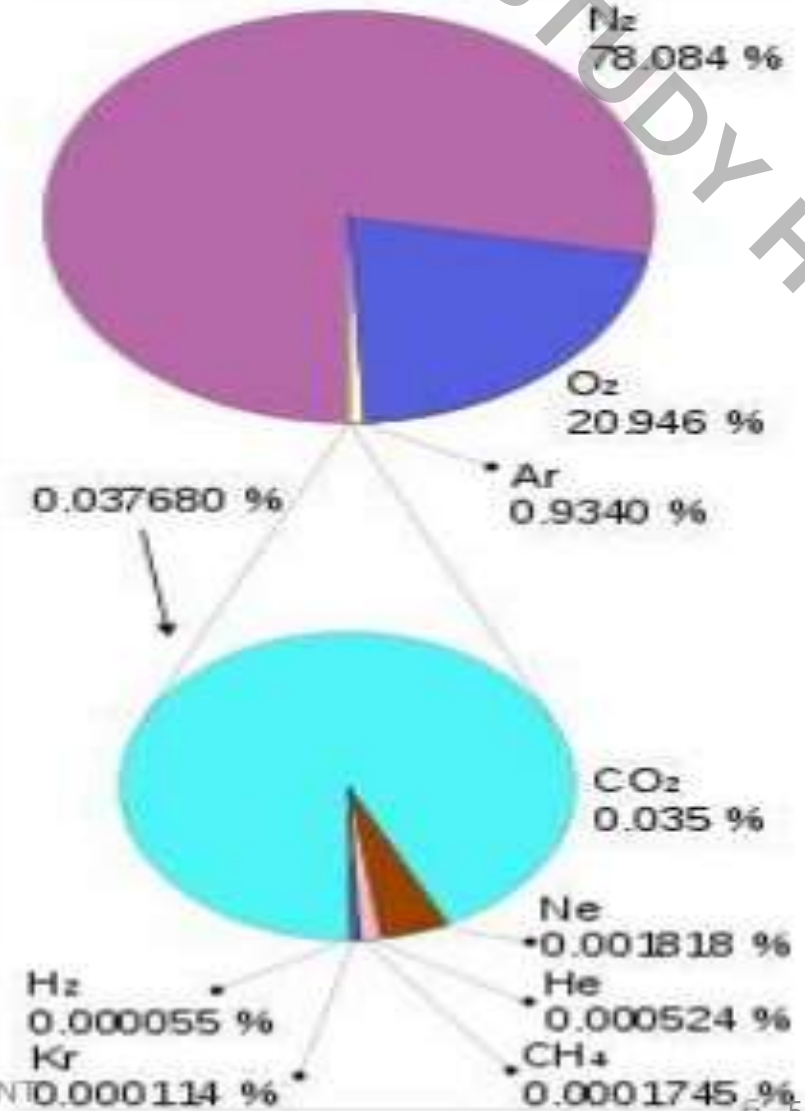
Cont....

- **Air: A mechanical mixture of Gases and Aerosols**
- **Pure air- colorless, odorless, tasteless and can't felt except in motion.**
- **By weight of 1 liter air= 1.3gm**
- **Pressure at sea level= 1033.6g/sq.cm (called 1 Atmosphere)**
- **Source Of O_2 – essential for life**
- **Source Of CO_2 – essential for plants in photosynthesis**
- **Without that would be – no clouds, no winds or storm and hence no weather.**

Composition of Atmosphere

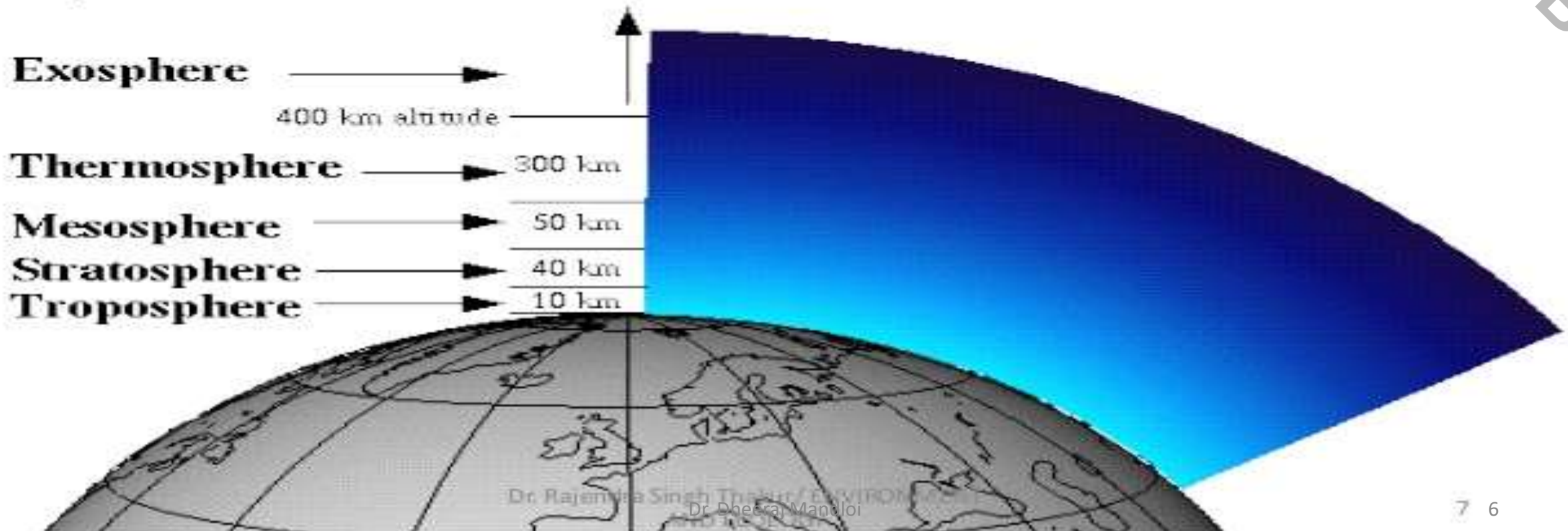
ppmv: parts per million by volume (note: volume fraction is equal to mole fraction for ideal gas only, see volume (thermodynamics))

Gas	Volume	%
Nitrogen (N ₂)	780,840 ppmv	(78.084%)
Oxygen (O ₂)	209,460 ppmv	(20.946%)
Argon (Ar)	9,340 ppmv	(0.9340%)
Carbon dioxide (CO ₂)	397 ppmv	(0.0397%)
Neon (Ne)	18.18 ppmv	(0.001818%)
Helium (He)	5.24 ppmv	(0.000524%)
Methane (CH ₄)	1.79 ppmv	(0.000179%)
Krypton (Kr)	1.14 ppmv	(0.000114%)
Hydrogen (H ₂)	0.55 ppmv	(0.000055%)
Nitrous oxide (N ₂ O)	0.325 ppmv	(0.0000325%)
Carbon monoxide (CO)	0.1 ppmv	(0.00001%)
Xenon (Xe)	0.09 ppmv	(9×10 ⁻⁶ %) (0.000009%)
Ozone (O ₃)	0.0 to 0.07 ppmv	(0 to 7×10 ⁻⁶ %)
Nitrogen dioxide (NO ₂)	0.02 ppmv	(2×10 ⁻⁶ %) (0.000002%)
Iodine (I ₂)	0.01 ppmv	(1×10 ⁻⁶ %) (0.000001%)
Ammonia (NH ₃)	trace	
Not included in above dry atmosphere:		
Water vapor (H ₂ O)	~0.25% by mass over full atmosphere, locally 0.001%–5% AND ECOLOGY	



Structure of Atmosphere

- **No sharp boundary** with extraterrestrial space
- **Phenomena** – magnetic and gravitational field extend outward for thousand km to vague zone of **nebular gases and radiation particle**.
- Basis of **Temperature and other related phenomenon** divided in to four major layer-

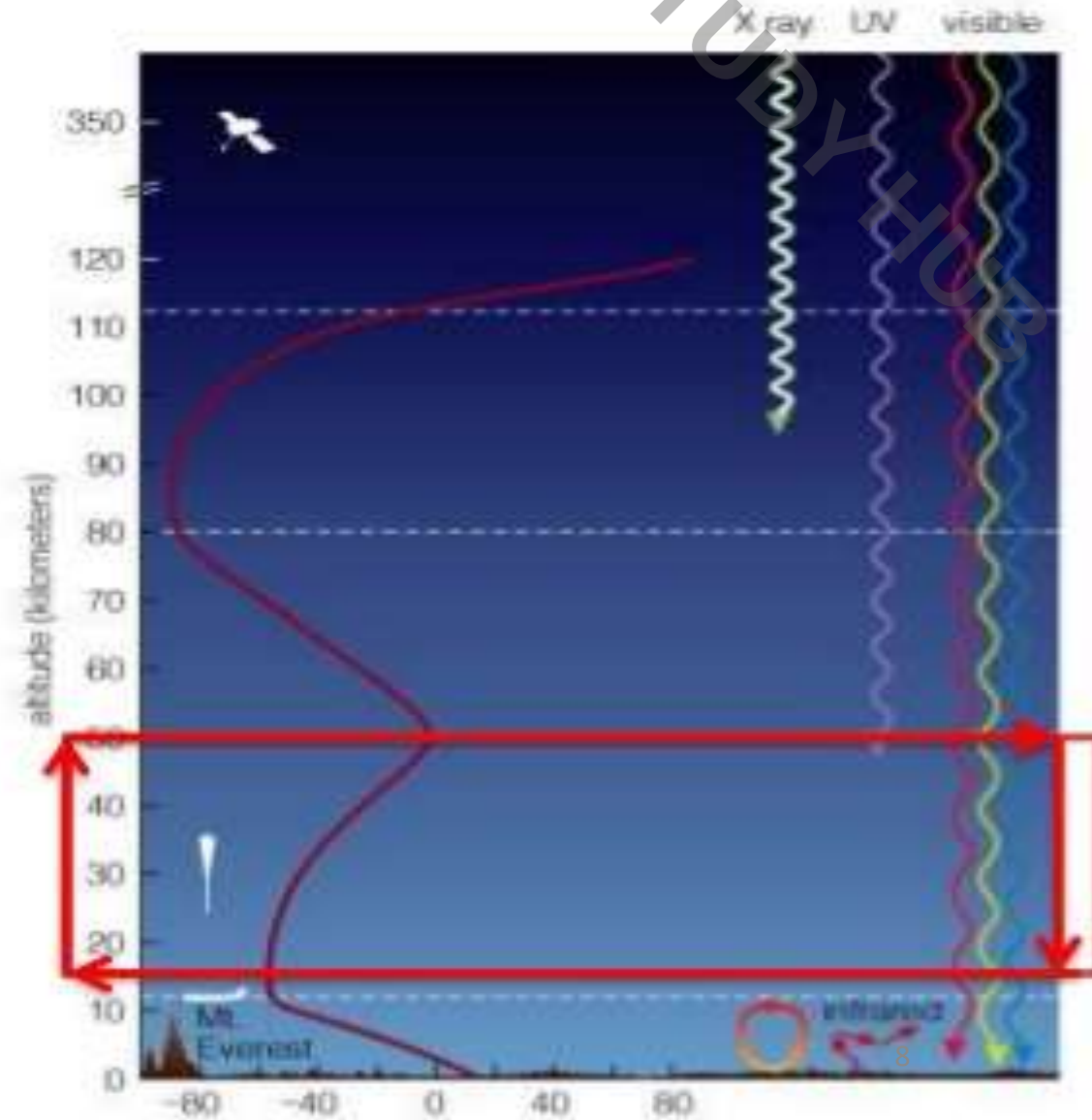


A spectral diagram showing the electromagnetic spectrum. The top part is a horizontal bar divided into three sections: 'X ray' (white), 'UV' (light blue), and 'visible' (rainbow colors). Below the bar, there are three vertical wave patterns: a white zigzag line for X-ray, a blue zigzag line for UV, and a rainbow-colored wave for visible light. A large diagonal watermark 'LET STUDY HUB' is overlaid on the image.

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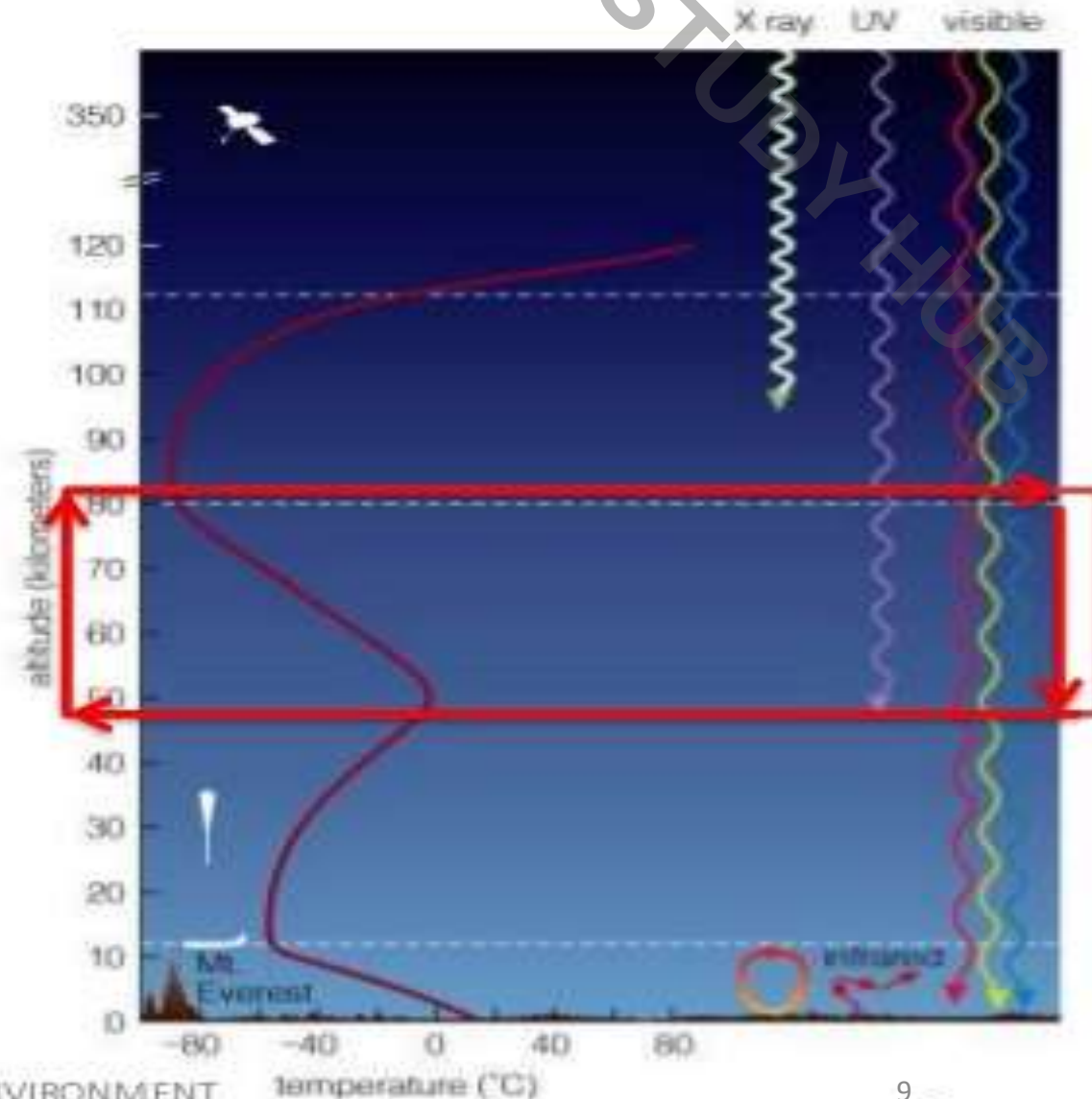
Stratosphere

- **16 to 50 km altitude**
- **Temperature increases with altitude and increases from -60°C up to a maximum of 0°C near outer limits, stratopause.**
- **Heating occurs because ozone (O_3) absorbs ultraviolet radiation from the Sun.**
- **Top of the stratosphere is known as the stratopause.**

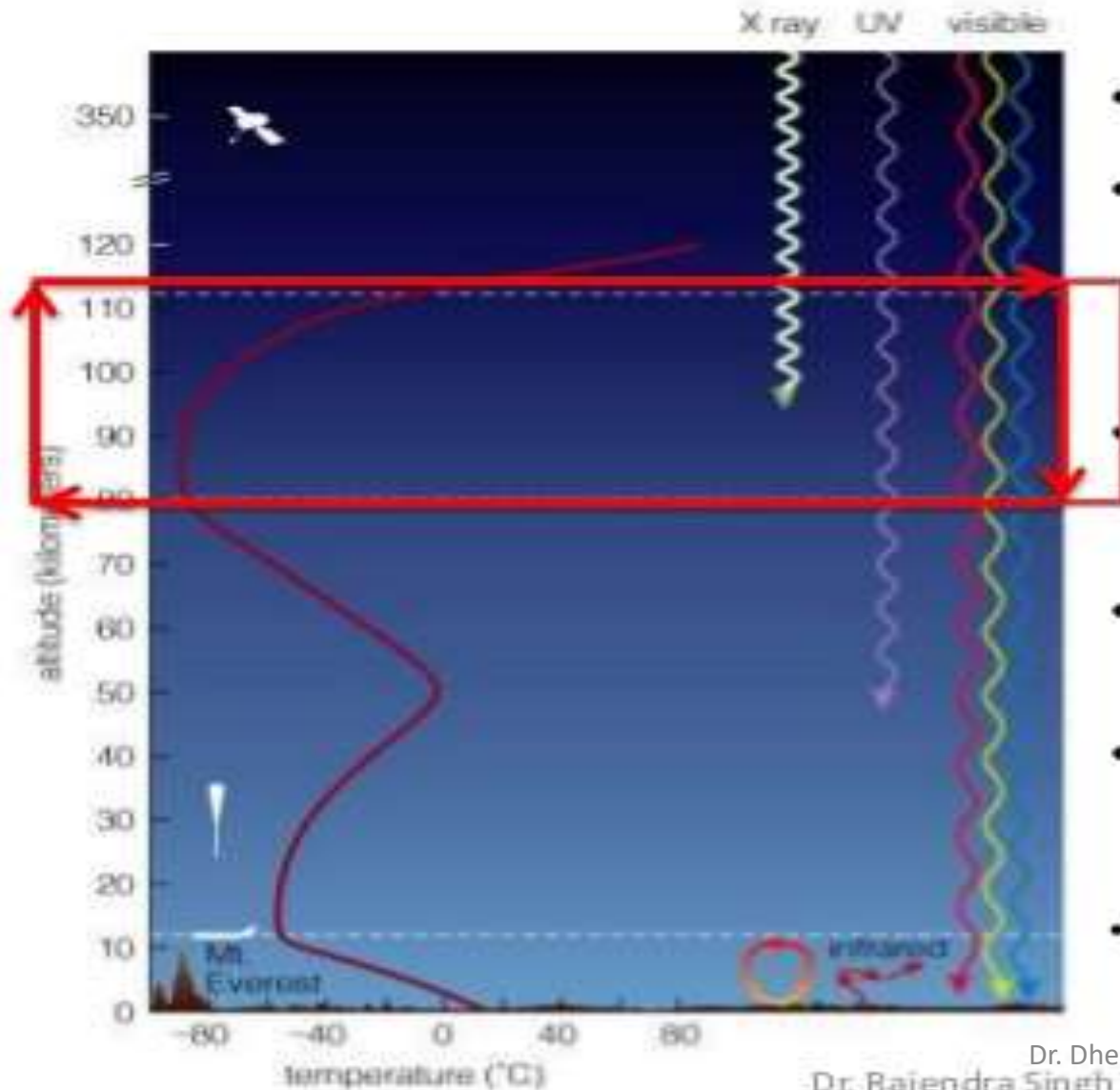


Mesosphere

- 50 to 80 km altitude
- Temperature decreases slowly with altitude **0°C to minimum about -75°C** near the mesopause.
- The **lowest temperatures in the entire atmosphere** are found at the mesopause during summer at high latitudes.
- Top of the mesosphere is known as the **mesopause at 80km**.

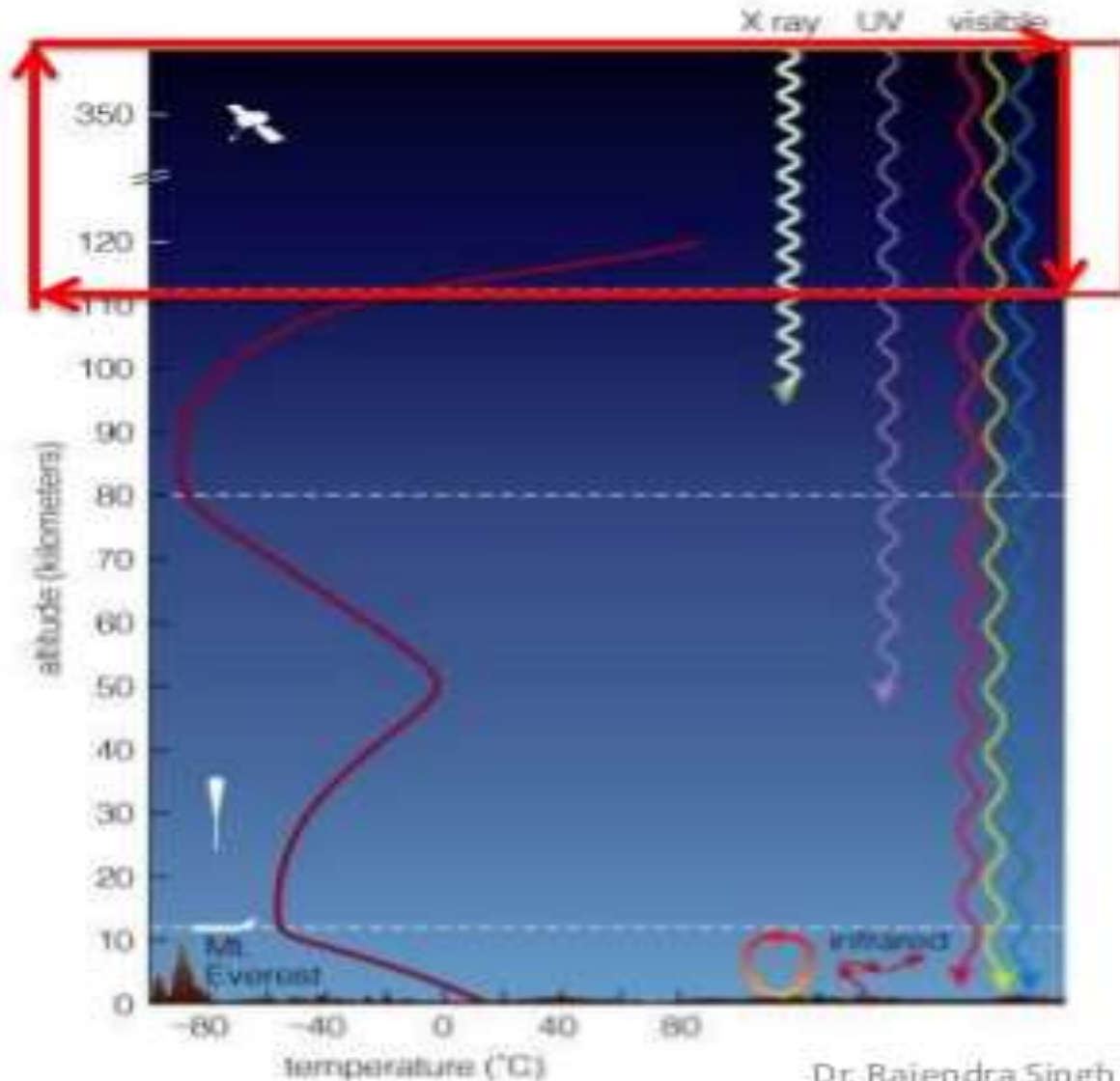


Thermosphere



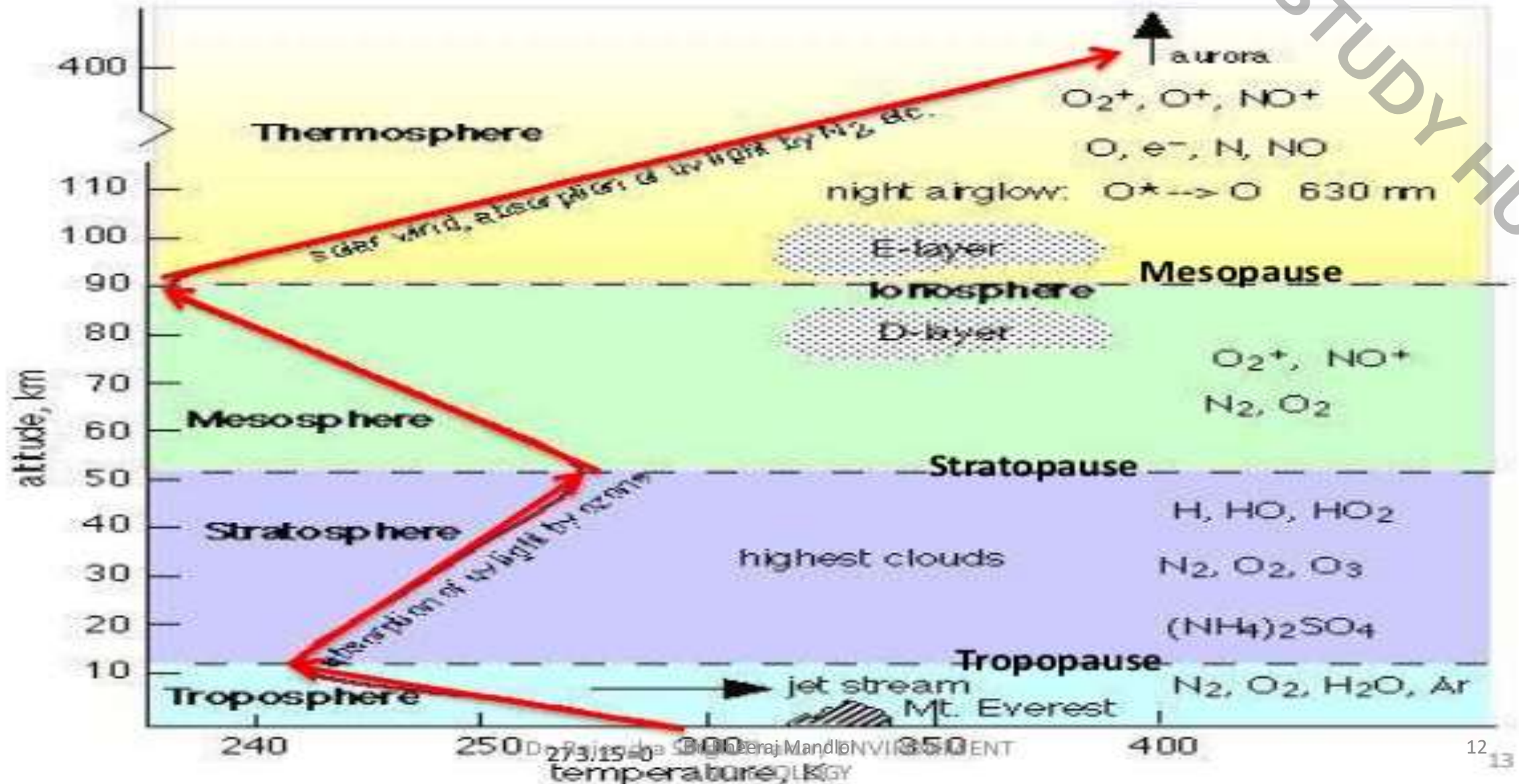
- **Thermosphere:** Layer at about 80 to 500 km altitude
- **Temperature increases** with altitude above 80 km.
- Additional vertical subdivision of atmosphere can be identified on the basis of chemical composition (ozonosphere) or physical properties other than temperature.
- Coinciding with the lower portion is called **Ionosphere** of the 100-400km.
- It is due to this layer that **radio waves are reflected by ionized layer** at great height.
- This heating is due to **absorption of solar radiation (wavelengths less than 0.2 microns)** by molecular oxygen (O_2).
- **X rays and ultraviolet light** from the Sun heat and ionize gases.

Thermosphere



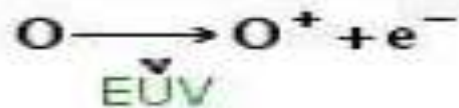
- **Exosphere:** Highest layer in which atmosphere gradually fades into space
- **Temperature rises** with altitude; **atoms** can escape into space
- The highest temperatures in the atmosphere can be found in the thermosphere, **2000°C** can occur.
- Warmed by **X rays and UV light**

Structure of Atmosphere

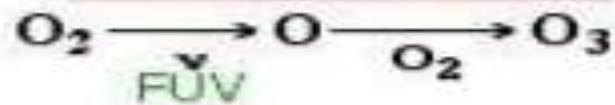


Photochemistry Examples

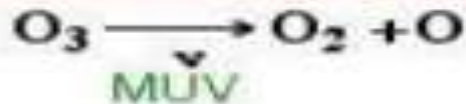
Creation of Ionosphere



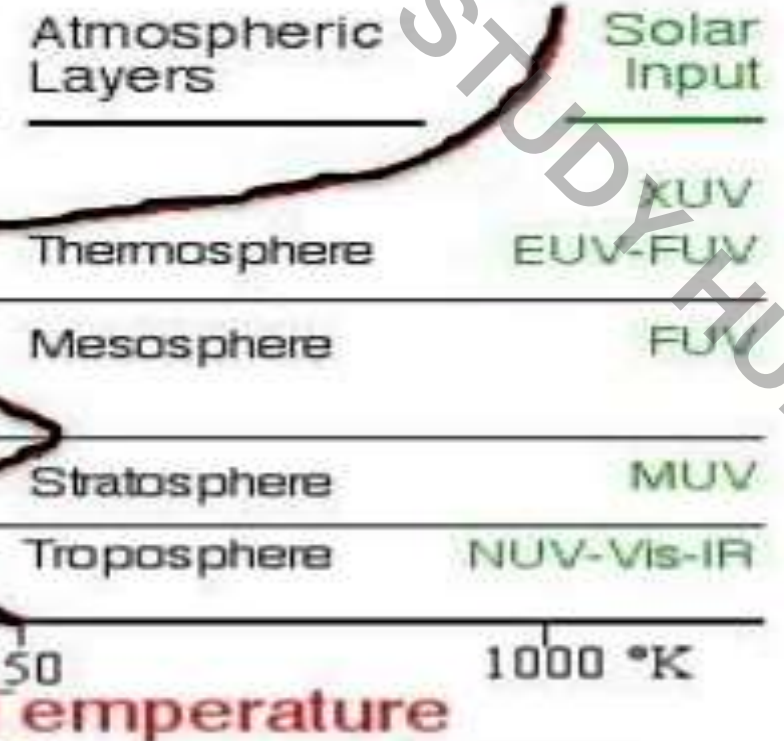
Ozone Creation



Ozone Destruction



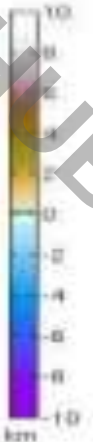
Heating



PLUS
other effects on
Climate
Surface
Oceans

2. HYDROSPHERE

- The hydrosphere includes **all water on Earth**.
- The abundance of water on Earth is a unique feature that **clearly distinguishes our "Blue Planet"** from others in the **solar system**.
- **Not a drop of liquid water** can be found anywhere else in the **solar system**.
- It is because the **Earth** has just **the right mass, the right chemical composition, the right atmosphere, and is the right distance from the Sun** that permits water to exist mainly as a liquid.



Hydrosphere

- The range of surface temperatures and pressures of our planet permit water to exist in **all three states**:
Solid (ice),
Liquid (water), and
Gas (water vapour).
- Most of the water is contained in the oceans and the high heat capacity of this large volume of water (1360 million cubic kilometres) buffers the Earth surface from large **temperature changes**.
- Water is the **universal solvent and the basis of all life on our Planet**.
- It is an **essential life-sustaining resource**.



Hydrosphere

- Water Resources
 - The hydrosphere consists of the Earth's water resources: **oceans, seas, lakes, rivers, streams, groundwater inflow with surface water, reservoir's and glaciers**
 - Earth's surface is **covered by 71% water**
 - Essential for life – can survive only a **few days without water**



The Structure of Hydrosphere

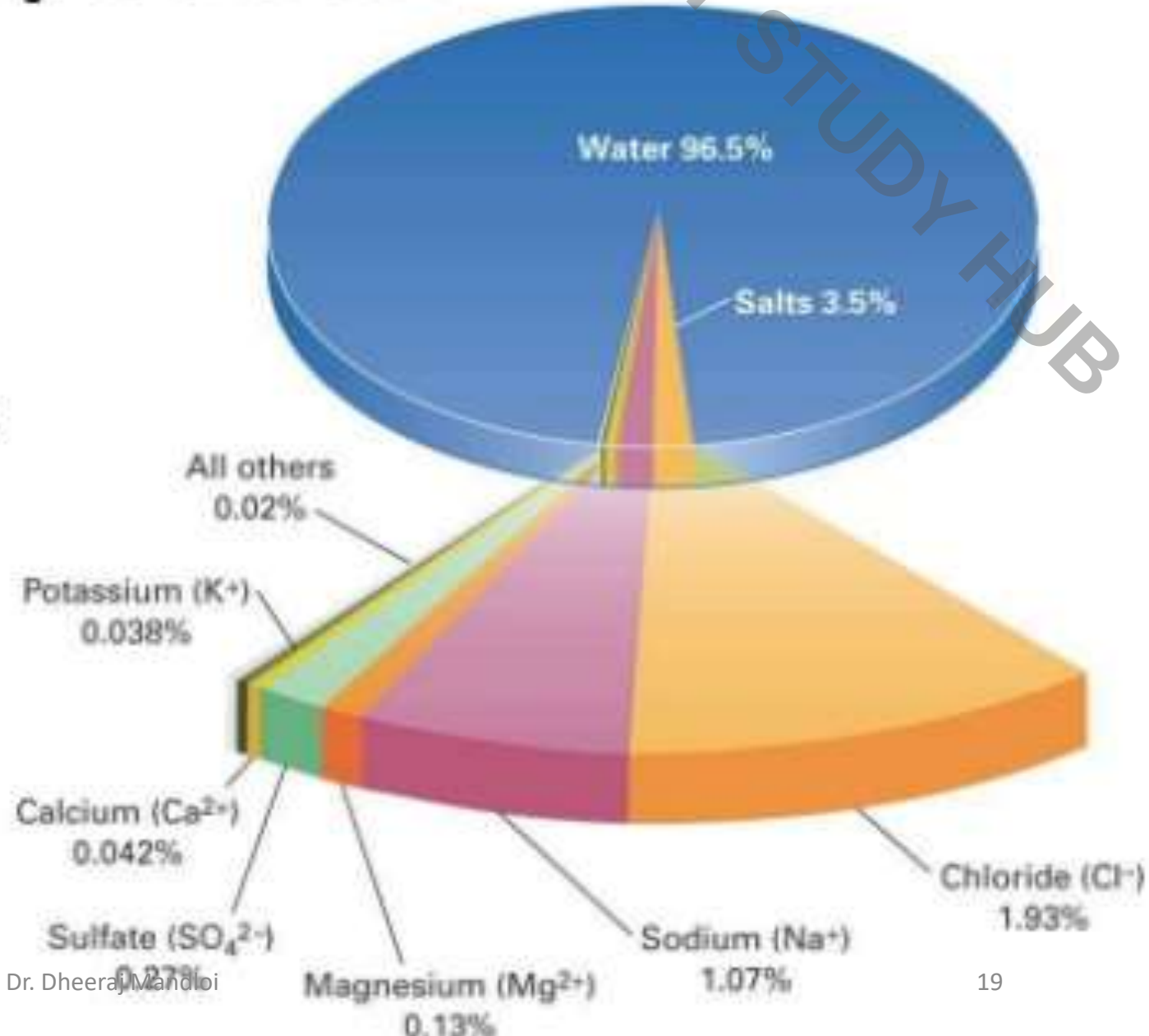
- **Oceans—97%** of water is **salt water** and it found in **the oceans**.
- **Fresh water**—the remaining **3%** is **freshwater**.
 - Fresh water distribution:
 - **Ice: 1.762%**
 - **Groundwater: 1.7%**
 - **Surface Fresh Water: 0.014%**
 - **Atmosphere and soil: 0.002%**

Water on Earth



Composition

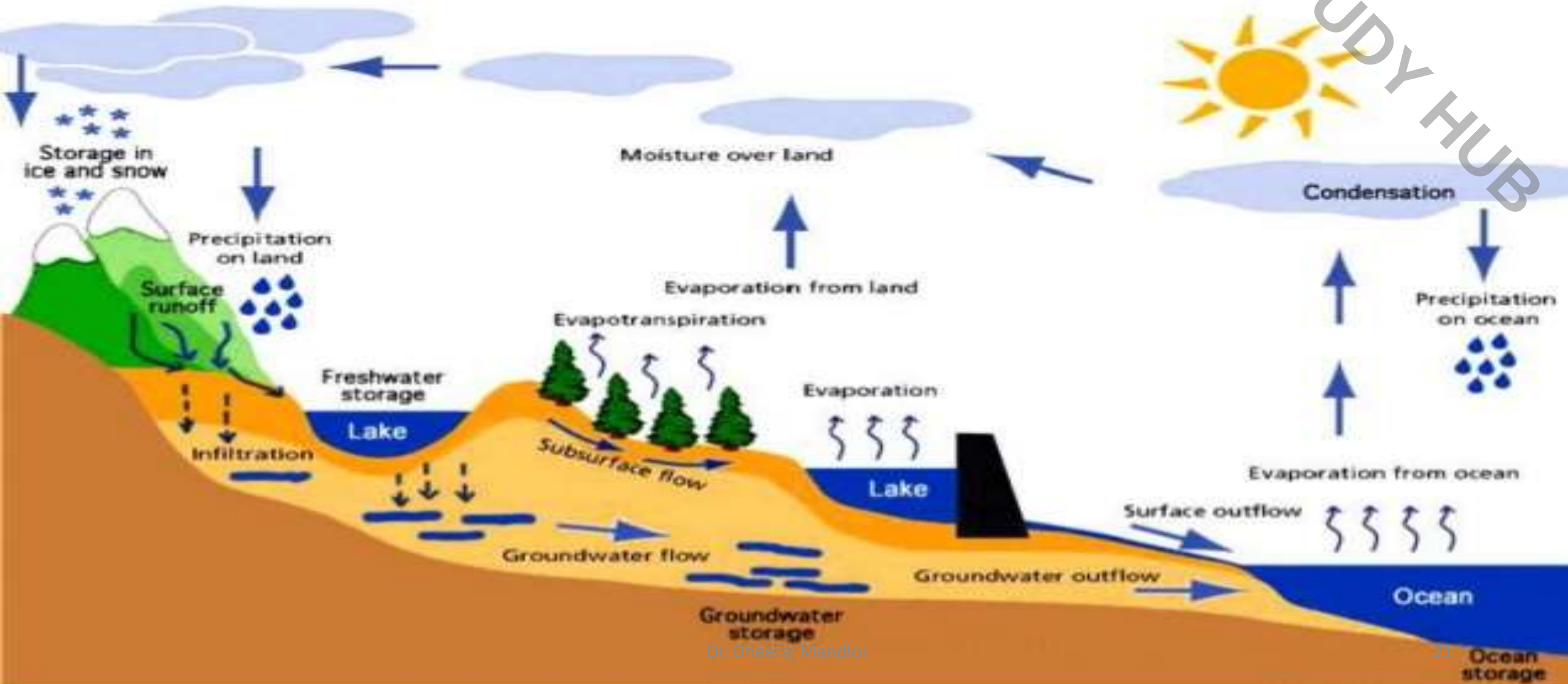
- ❑ Obviously water
- ❑ But also:
 - Dissolved minerals
 - Dissolved gases
 - Particulates



WATER CYCLE

- The **continuous movement of water into the air, onto land and then back to water** sources is known **“Water cycle”**.
- **Evaporation** is the process by which as **liquid water** is heated by the sun and **then rises** into the atmosphere as **“water vapour”**.
- Water continually evaporates from **Earth’s oceans, lakes, streams, and soil**, but the majority of the water evaporates from the oceans.
- In the process of **Condensation**, **water vapour** forms **water droplets on dust particles**.
- These **water droplets form clouds**, in which the **droplets collide, stick together, and create larger, heavier droplets**.
- These larger droplets fall from clouds as rain in the process called

Water Cycle



3.LITHOSPHERE

- The lithosphere is **the solid, outer part of the Earth.**
- Geological speaking, **top crust of the earth** on which the **continent and ocean basin rest.**
- The Earth consists of **three main layers: the core**, or the inner layer; **the mantle**, in the middle; and **the crust**, which includes **the continents and ocean floor.**
- Thickest in continental regions where **average thickness 40km** and **thinnest in ocean 10-12km.**
- The movement of the lithosphere, called **plate tectonics**, is the reason behind a lot of Earth's most dramatic geologic events.
- When **one plate moves beneath another**, or **when two plates rub together**, they **can create earthquakes and volcanoes.**

- The earth is **divided into three layers –the crust, mantle, and core-based** on what each one is made of- **The lightest materials make up the outermost layer, and the densest materials make up the innermost layers.**
- The **core is approximately 33% of the Earth's mass** and is the innermost layer.
- The **mantle is approximately 67% of the Earth's mass.**
- The outermost layer is the **crust approximately 1% of the Earth's mass.**
- Environmental scientist- **interest upper few feet of soil.**
- **Soil** which is important part for **organic matter and biological activities**
- To produce food for **human being & animals** but also **decomposition of organic waste** is carried out by a host of **micro-organism in the soil.**

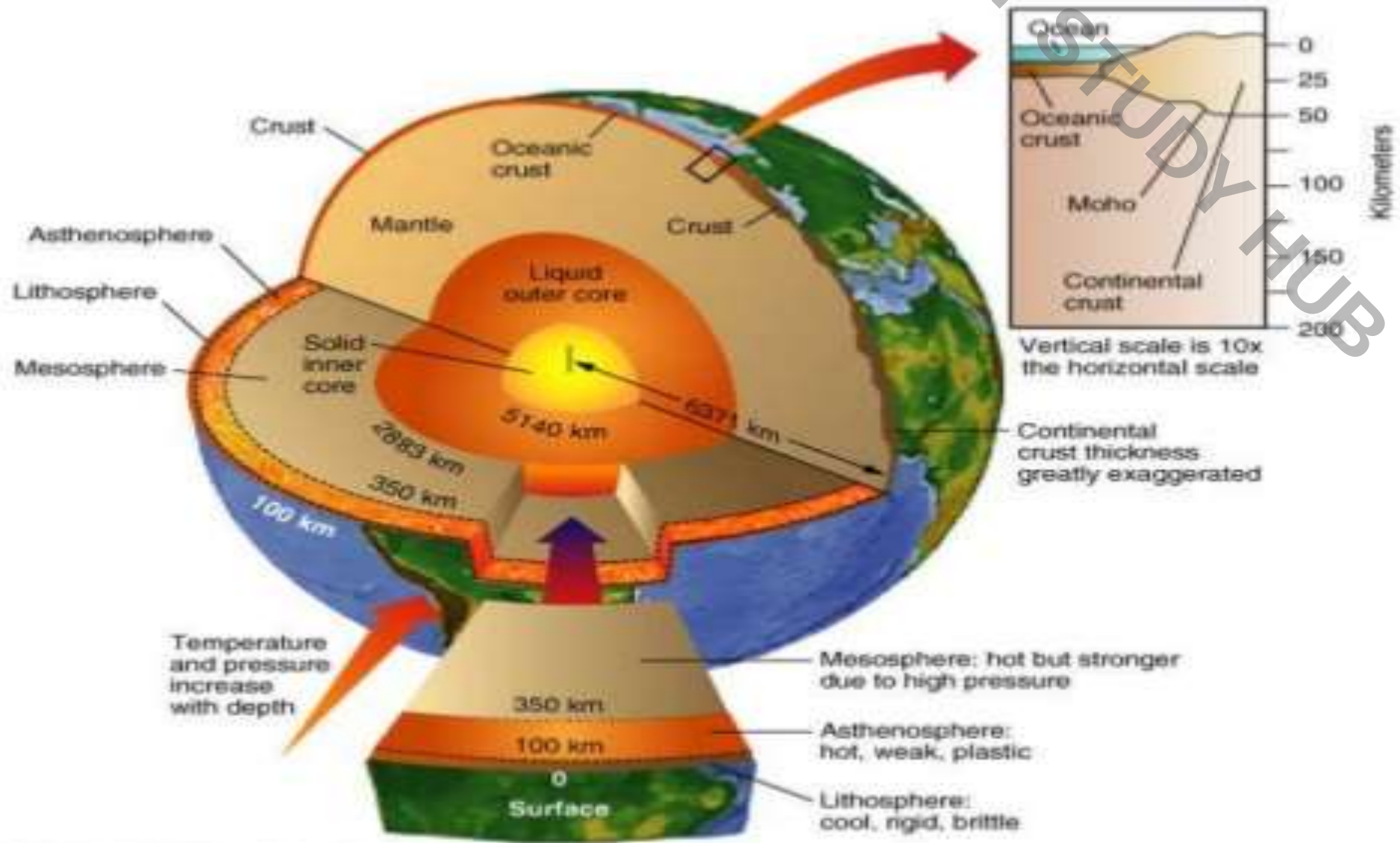
Earth structure: The main units

Compositional:

- Crust
- Mantle
- Core

Rheological:

- Lithosphere
- Asthenosphere
- Mesosphere



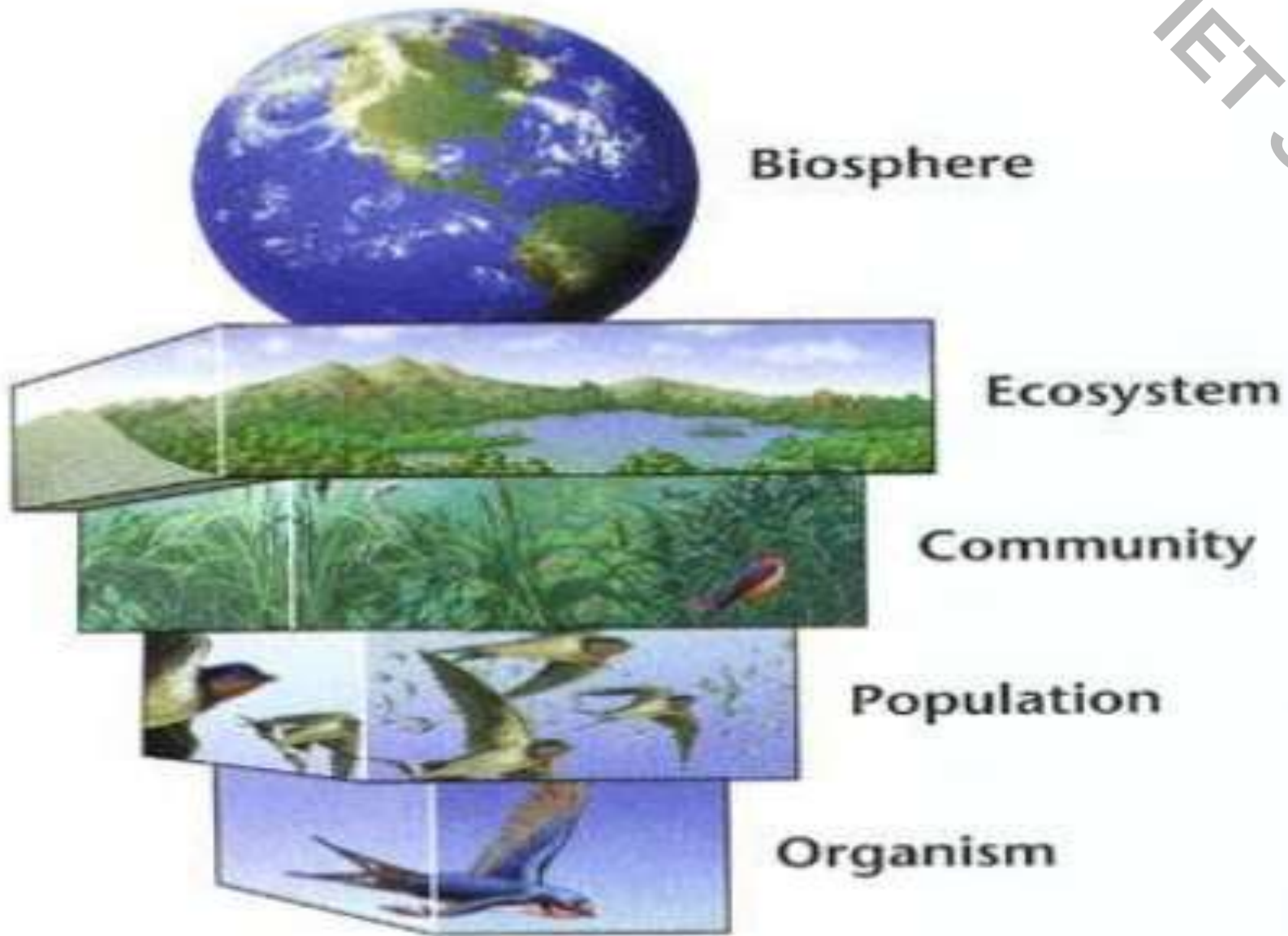
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4.BIOSPHERE

- The interaction of life with environment takes place of many levels.
- A single bacterium in the soil interact with the water, air and particle of soil within a fraction of a cubic centimeter.
- While forest extending hundred square km interact with large volume of water, air and soil.
- Part of Earth in which life exists including land, water, and air or atmosphere.
- We live in the natural world and use its resources (water, space, food, etc).
- The natural world effects our lives (weather, fire, economy).
- To protect biodiversity.
- Thus all lower atmosphere and near surface part of lithosphere and hydrosphere affected by life is whole exist few meters of exists surface.
- This region of the earth where life exist is known as “Biosphere”.
- All living things required energy and materials.
- In biosphere energy received from the sun and interior of the earth received energy is then used and given off a material are recycled.

Biosphere

- The biosphere is a **thin shell that encapsulates the earth** which includes **all life as well as the lower atmosphere and the oceans, rivers, lakes, soils and solid sediments** that actively interchange materials with life.
- According to an estimate biosphere contains-
 - ✓ More than **3.5 lacs species of plant** (including-algae, fungi, mosses & higher from of plants).
 - ✓ More than **11 million species of animal** (ranging from **unicellular protozoa to man**).
- **Supplies all the essential requisite of life-namely, light, heat, air, water, food & living space (habitats) for all these species.**
- Since biosphere is **very large and complex** then it better understanding by **divided in to smaller units called “Ecosystems” or “Ecological systems.”**



Describe role of various Natural resources in Engineering and development. (including their role/ impact in our life)

NATURAL RESOURCES

Resources provided by Nature including Forests, Water, Minerals, Food, Cultivars, Energy, Land etc.

Include our interaction with these natural resources and what role they play in our life and economy.

What is Natural Resource?

Definition/Meaning:

The environment is everything which surrounds an organism and influences its life in many ways. It includes physical and biological components. The physical components of the environment are soil, water, air, light and temperature (Abiotic components). The plants and animals are collectively referred to as Biotic components. All these components work together, interact and modify the effect of one another. The basic needs of life are fulfilled by minerals present in nature. These are referred to as Natural Resources.

Importance of Natural Resources

- Least wastage and maximum economic advantage.
 - Availability of natural resources at minimum transportation cost.
 - Selection of industrial zone.
 - Integral use of Natural Resources to obtain long term Economic advantages. (eg., proper plan to utilize timber from forest for long period of time)
- To achieve multipurpose advantages. (eg., dam construction on a river serves for protection against the flood, proper irrigation project, development of fishing industry, electricity generation.

Classification of Natural Resources

1. Based on Origin:

- a. BIOTIC: (living organisms)
- b. ABIOTIC: (non living organisms)

2. Based on Availability:

- a. Inexhaustible: (replenishes naturally)
- b. Exhaustible: (replenishing process is extremely slow)

3. Based on Distribution:

- a. Omni present: (found everywhere)
- b. Location based: (found at some specific locations only)

Continue...

4. Based on Source:

- a. Water Resources
- b. Forest Resources
- c. Mineral Resources
- d. Marine Resources

5. Based on Chemical Composition:

- a. organic: (vegetables, animals, bacteria, mineral oil)
- b. Inorganic: (Air, water, minerals)
- c. mixed: (land)

Importance of Land in India:

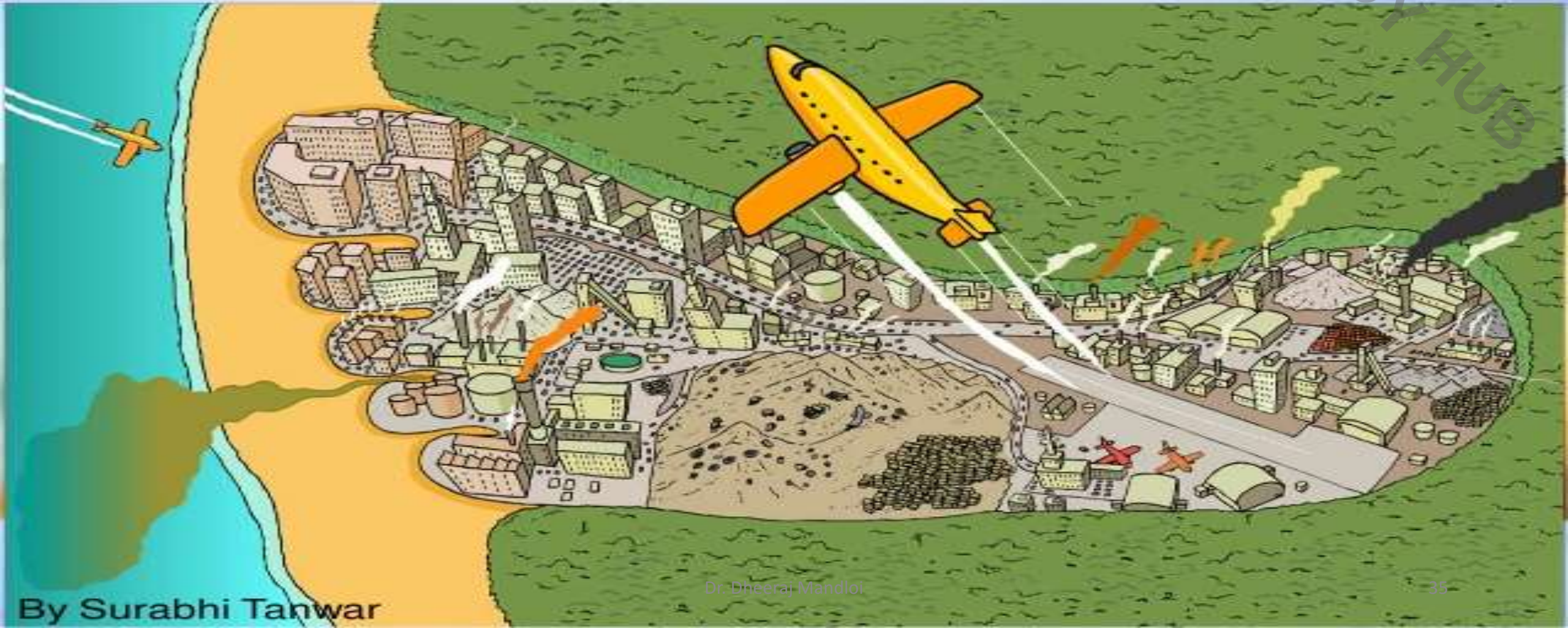
- India is an Agricultural Economy and it is inevitable to have Agricultural activities without proper usage of available land resource.
- 60% of the land in India is either cultivable or non cultivable in which $\frac{1}{3}$ of which is not proper for crop-yield due to soil erosion and less mineral content in the land.
- $\frac{1}{3}$ land is having less productivity.
- So we have only $\frac{1}{3}$ land available for crop yield.
- Due to deforestation the problem of soil erosion is increasing day by day.
- To increase the crop production, we have to use scientific farming techniques and will have to look for the solutions towards soil erosion.

Steps to conserve the Land:

1. Waste land reclamation. (convert the non cultivable land into cultivable land)
2. Adopting diversity by the use of proper irrigation technique.
3. The Agricultural lands should be kept reserved for the Agricultural use only.
4. Use of Hybrid seeds should be promoted to maintain the fertility of the land for long period.
5. Grow more trees to avoid soil erosion.
6. Farmers should be given advices after proper testing of the land minerals.

Adverse Impacts of Development on our Environment

Impact of Development On Environment



Industry

Economy, Conveniences and Productivity advances

Industrial
Revolution
mid 1800's.

● More "Stuff" → More Waste

Increase use of Tools, Technology and Sciences

Use of Fossil Fuels → Release of CO₂ → Green House Effect

Chemical and waste by products of manufacturing

Air, Water, Land, Noise Pollution

Including-

Definition

List and classification of pollutants with examples

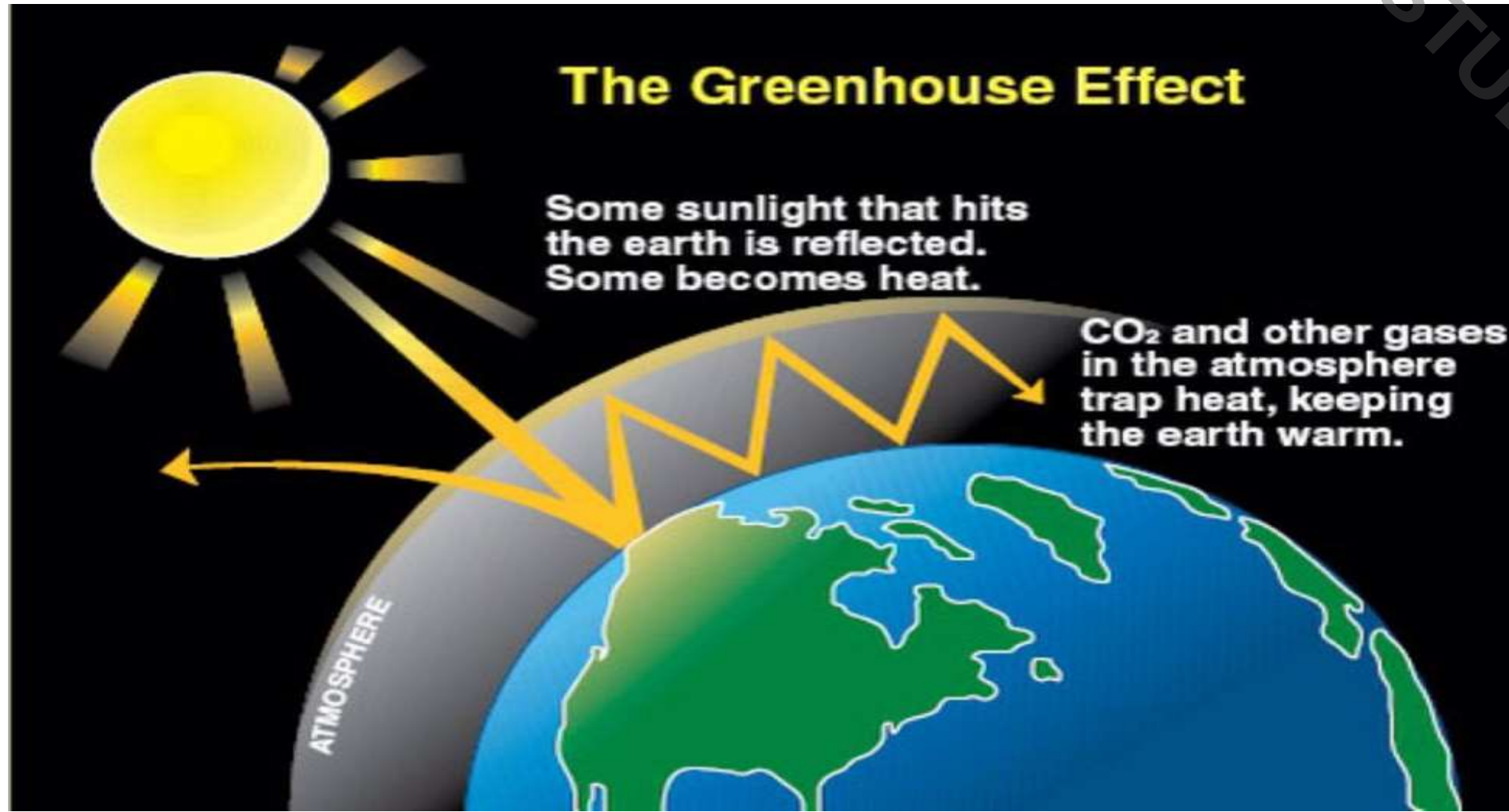
Their sources

Harmful effects

Control measures

IET STUDY HUB

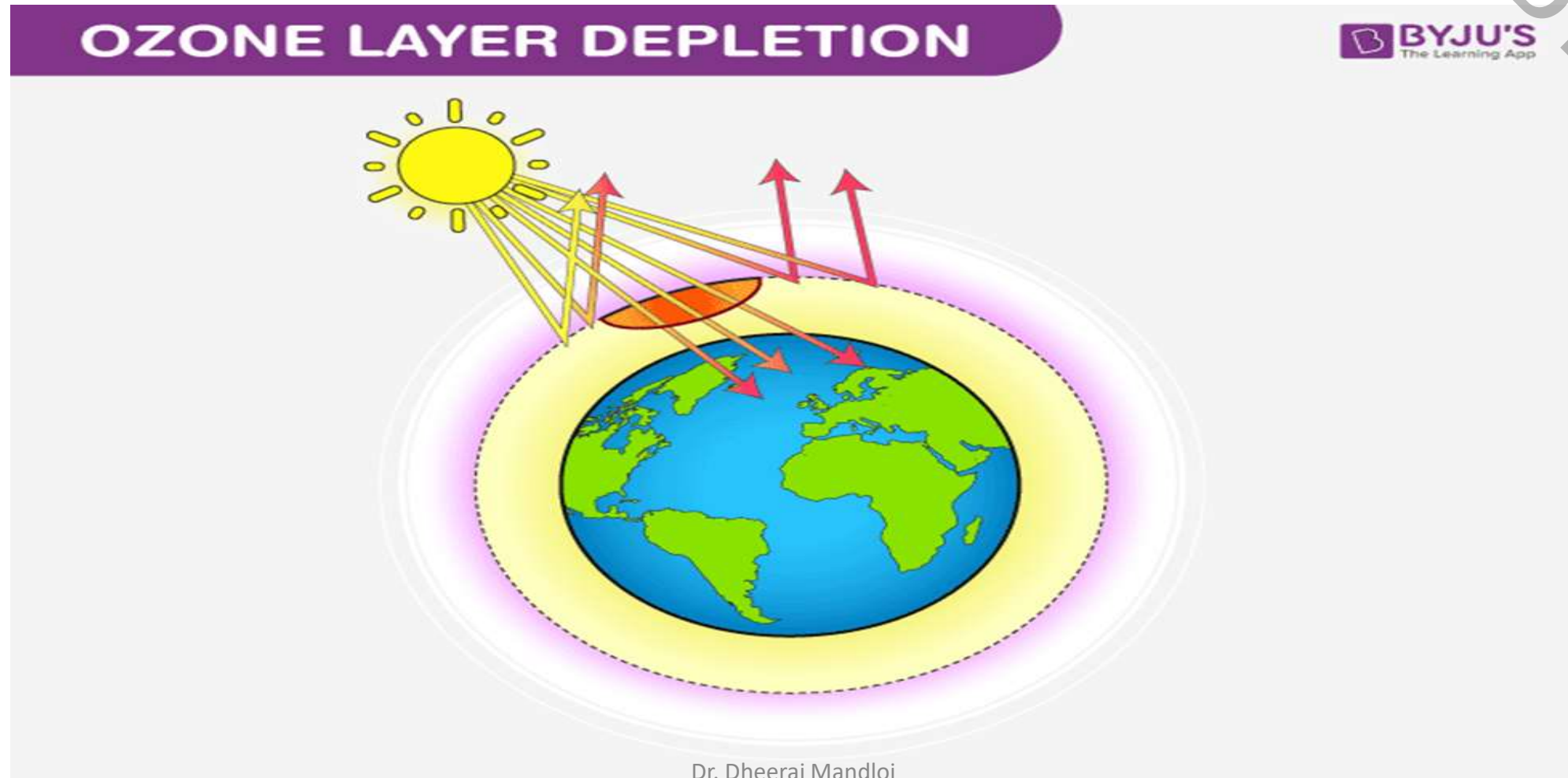
Green House Effect and Global Warming



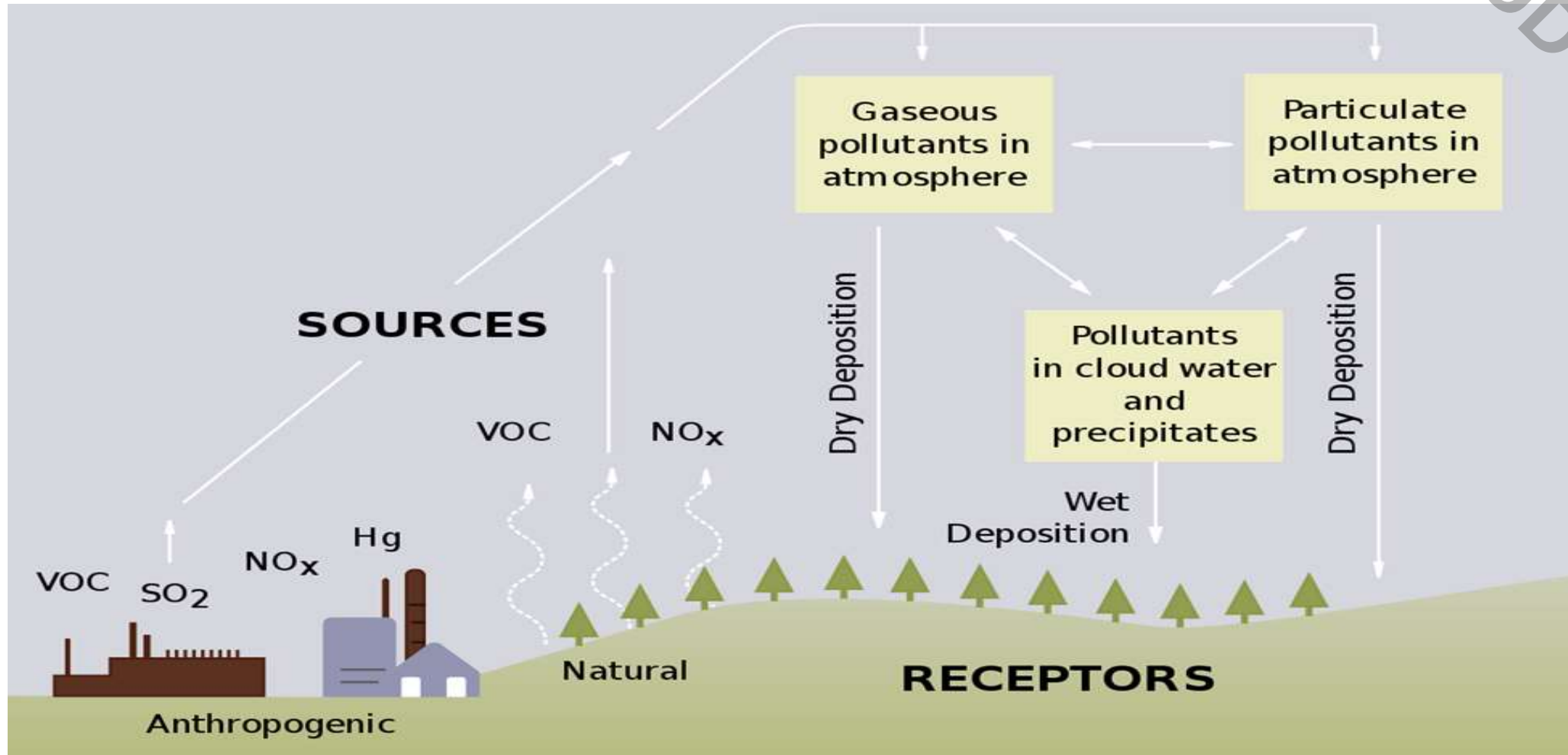
Explain concept of Green house effect and Global warming, green houses gases are increasing (why?), effects, control measures



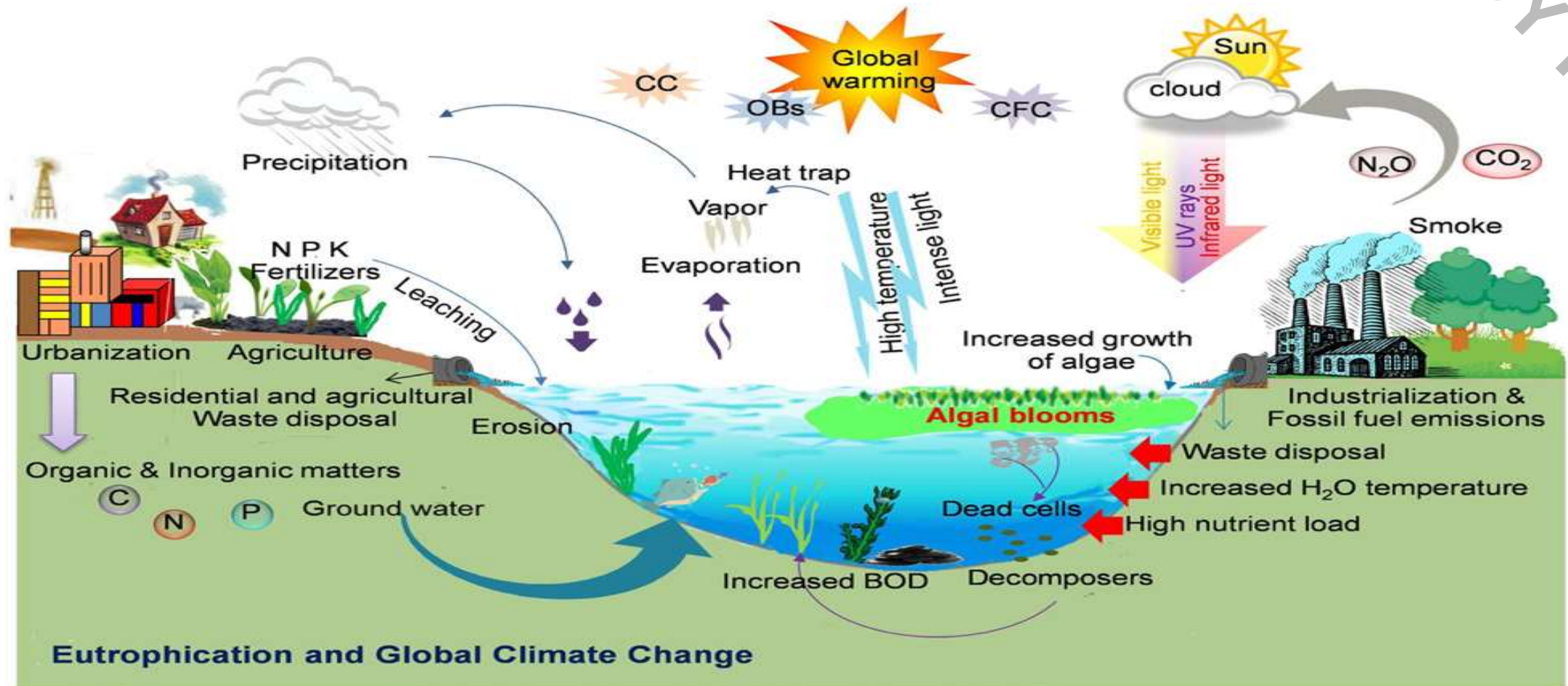
Chlorinated compounds are present in very small amount but causing maximum damage to Ozone layer in stratosphere. Why and how? (Chain reactions, more life of $\text{Cl}\cdot$)



What causes Acid rains and what are its adverse effects to our environment? (SO_x , NO_x , sources, effects, control)

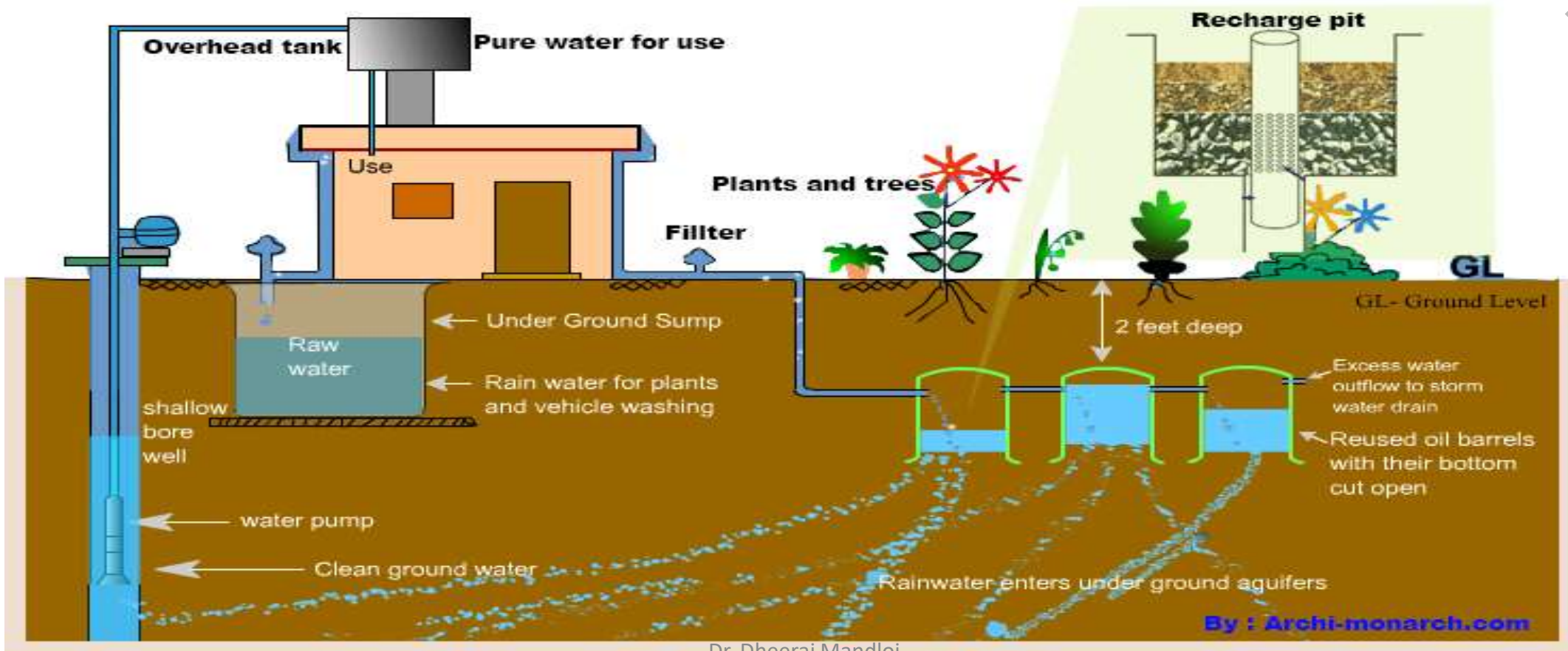


What is Eutrophication? What factors are responsible for this? Why this is considered very serious problem associated with water bodies? (gradual death of a water body)



Why Rain water harvesting is essentially required in present times? Explain various methods available for Rain water harvesting. (with examples)

Awareness, methods of rain water harvesting, its importance



What is Environment Impact Assessment? What steps are involved in EIA? How it is helpful in environment protection and pollution control?

EIA is considered as one of the most effective engineering and management tool to control pollution.

Now, mandatory in India also.

It includes prior assessment of various adverse impacts on environment before start of a project.

Need to compensate these adverse effects and then proceed with the project.

Environment Impact Assessment in India

Environment Impact Assessment (EIA) is a formal process used to predict the environmental consequences of any development project. Environment Impact Assessment in India is statutory backed by the Environment Protection Act in 1986, which contains various provisions on EIA methodology and process.



Rationale behind EIA

EIA looks into various problems, conflicts and natural resource constraints which may not only affect the viability of a project but also predict if a project might harm to the people, their land, livelihoods and environment. Once these potential harmful impacts are predicted, the EIA process identifies the measures to minimize those impacts. Thus, the objective of the EIA is to:

- Identify the environmental, social and economic impacts of a project prior to taking a decision on its implementation.
- Mitigation of harmful impacts and maximizes the beneficial effects.

Once the assessment is complete, the EIA findings are communicated to all stakeholders viz. developers, investors, regulators, planners, politicians, affected communities etc. On the basis of the conclusion of EIA process, the government can decide if a project should be given environment clearance or not. The developers and investors can also shape the project in such a way that its harms can be mitigated and benefits can be maximized.

Historical Facts

The EIA process finds its origin from United States where due to huge public pressure; the government enacted National Environmental Policy Act (NEPA) in 1970s. The role of EIA process was formally recognized at the Earth Summit in Rio Conference in 1992 in which the Rio declaration stated that EIA

Explain the concept and need for Sustainable development.
(including Smart cities, local is vocal, economy, health, education, environment, good governance).

5 R – Reduce, Reuse, Recycle, Residue disposal, Research.
Take at least one example of each R





SUSTAINABLE DEVELOPMENT GOALS





